

MuTau channel: Combine tool

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Overview

Goal

Set a 95% CL upper limit on the signal strength $r = \sigma_{\text{obs}}/\sigma_{\text{SM}}$ for the exclusive $\gamma\gamma \rightarrow \mu\tau$ process using the CMS Combine tool.

Full pipeline

```
save_shapes.cpp      mutau_mass_shape.txt      combine
(shape histograms) --> (datacard + workspace) --> (AsymptoticLimits)
      |                                     |
MuTau_shapes.root    higgsCombineMuTau.AsymptoticLimits.mH120.root
                        |
                        plot_single_limit.py
                        |
                        MuTau_limits.png / limits.json
```

Step 1: save_shapes.cpp

Produces `MuTau_shapes.root` – the input ROOT file with per-sample invariant mass histograms that Combine reads as shapes.

Input samples

Sample	File
Data	<code>Data_2018_UL_MuTau_nano_merged_proton_vars.root</code>
DY	<code>DY_2018_UL_MuTau_nano_merged_pileup_protons.root</code>
ttbar	<code>ttjets_2018_UL_MuTau_nano_merged_pileup_protons.root</code>
QCD	<code>QCD_2018_UL_MuTau_nano_merged_proton_vars_new.root</code>
Signal	<code>MuTau_sinal_SM_2018_july.root</code>

Step 1: save_shapes.cpp

What it does

- ① Loops over all events; requires `sist_mass > 0`
- ② For MC: requires protons on both arms (`xi1 > 0 && xi2 > 0`)
- ③ Fills invariant mass histogram `m_X` for each sample (30 bins, 0-1200 GeV)
- ④ Also fills 7 other kinematic histograms per sample (`acop`, `sist_pt`, `sist_rap`, `tau_pt`, `met_pt`, mass matching, rapidity matching)
- ⑤ Applies MC normalization after the loop:
 - DY: $\text{Scale}(1.004\text{e-}4) = L \times \sigma / \sum w$
 - ttbar: $\text{Scale}(1.0)$ (already normalized via 0.15 factor in Phase1)
 - Data, QCD: $\text{weight} = 1.0$ (data-driven)

Output

- `MuTau_shapes.root` containing `m_data`, `m_sinal`, `m_dy`, `m_ttjets`, `m_qcd`

Step 2: datacard (mutau_mass_shape.txt)

Channel and process structure

```
imax 1    # 1 channel: mutau
jmax 3    # 3 backgrounds: DY, ttbar, QCD
kmax *    # all nuisance parameters
```

Shape mapping (all from ../MuTau_shapes.root)

Process	Histogram	Index
data_obs	m_data	–
sinal	m_sinal	0 (signal)
dy	m_dy	1
ttjets	m_ttjets	2
qcd	m_qcd	3

All rates set to -1 -> Combine reads normalization directly from histogram integrals.

Step 2: systematic uncertainties

Lognormal (lnN) uncertainties

Source	Signal	DY	ttbar	QCD
Luminosity	1.5%	1.5%	1.5%	–
tau ID vs jet	5.0%	4.0%	4.0%	–
tau ID vs e	2.5%	2.5%	2.5%	–
tau ID vs mu	3.1%	3.2%	2.6%	–
Muon trigger	0.5%	0.07%	0.07%	–
Muon ID+Iso	0.02%	0.04%	0.04%	–
Muon reco	0.06%	0.05%	0.05%	–
Proton xi	50%	50%	50%	50%

Floating normalizations (rateParam, range [0, 5])

- beta_dy for DY, constrained by dy_norm lnN 1.30
- beta_tt for ttbar, constrained by ttjets_norm lnN 1.50

Step 3: running

Commands

Convert datacard to RooWorkspace

```
text2workspace.py datacards/mutau_mass_shape.txt \  
  -o datacards/mutau_mass_shape_workspace.root \  
  -m 120
```

Compute asymptotic limits at 95% CL

```
combine -M AsymptoticLimits datacards/mutau_mass_shape_workspace.root \  
  -m 120 -n MuTau --cl 0.95
```

Output file

higgsCombineMuTau.AsymptoticLimits.mH120.root

quantileExpected	Meaning
-1	Observed
0.5	Expected median

Step 4: plot_single_limit.py

purpose

Reads the output and produces a summary text file and limit plot.

sage

```
python3 datacards/plot_single_limit.py \  
    higgsCombineMuTau.AsymptoticLimits.mH120.root \  
    --mass 120 --output datacards/MuTau_limits
```

what it does

- ① Reads quantiles from the limit tree
- ② Writes MuTau_limits.txt – plain-text table of all six values
- ③ Draws a TMultiGraph:
 - Yellow band: $\pm 2\sigma$ expected
 - Green band: $\pm 1\sigma$ expected
 - Dashed black line: expected median
 - Black point: observed

Results: 95% CL upper limits on r

method: AsymptoticLimits, $m_H = 120$ GeV

Estimate	Limit on $r = \sigma/\sigma_{\text{SM}}$
Expected -2σ	3156
Expected -1σ	3987
Expected (median)	4000
Expected $+1\sigma$	8050
Expected $+2\sigma$	12131
Observed	13891

The large values of r indicate the analysis is not yet sensitive to the SM signal – the current selection and statistics allow excluding signal cross sections $\sim 4000\times$ the SM prediction at 95% CL.

Results: limit plot

